

Project Work Plan for Research and Data Integration for Cancer through Advanced Linkage “RADICAL”

Sponsored by American College of Radiology

Funding provided by Advanced Research Projects Agency for Health (ARPA-H)

Background

Medical images represent an invaluable source of clinical information critical to answering research questions and developing AI and advancing precision medicine. But barriers to creating centralized image repositories have inhibited the use of the vast majority of imaging performed across the imaging community. In response, the American College of Radiology (ACR) is leading efforts to develop approaches to promote use of images which remain within the data ecosystem of the imaging facility. The RADICAL project will contribute to ACR’s vision by demonstrating a distributed architecture (Appendix A) that enables the creation of federated datasets comprised of medical imaging and clinical data that is then used to test an AI use case.

AI training and validation is hindered by the lack of availability of diverse and up-to-date data sets. In addition, the process of navigating individual site data transfer agreements is challenging and time consuming. This process also requires storage of redundant data external to the providing sites. A distributed network of diverse indexed data offers a more efficient and robust approach to AI testing and validation. In this approach indexed data remains internal to the sites, and AI testing and validation is performed “in house” with only the algorithm results exported for central aggregation and analysis. From a vendor’s perspective, this is more efficient as it requires contracting with only a single entity to access data and targeted data sets can be requested to suit the needs of individual projects. In this project, the distributed network approach will be piloted with at least one participating imaging center, demonstrating the feasibility of privacy-preserving record linkages between imaging datasets and external data sources. The second objective is to demonstrate the application of AI algorithms on linked imaging data to generate new analytic value. Site AI algorithm results will then be aggregated and analyzed centrally at the ACR. These two workstreams are related but not dependent, allowing each to be evaluated on its own merits and to further future research while preserving site autonomy and patient privacy. Site AI algorithm results will be aggregated and analyzed centrally at the ACR.

Project Aims

Our aims for the RADICAL Project (Research and Data Integration for Cancer through Advanced Linkage) include the following:

- Leverage privacy preserving linkages (PPRL) to demonstrate the ability to create distributed “virtual” repositories consisting of medical images retained within the imaging facility and external data sources;
- Demonstrate the value of a distributed AI network approach to testing and validating AI algorithms by leveraging a breast cancer use case via an AI algorithm to generate novel analytic value;
- Apply ACR medical imaging indexing methodology (Appendix B) to a cohort of medical images retained at local imaging sites and create a dataset of indexed imaging studies;
- Demonstrate support of Medical Imaging and Data Resource Center’s (MIDRC’s) objective to create a national library of indexed medical imaging studies by enabling the breast cancer-related imaging study indexed by ACR to be incorporated into the MIDRC library of indexed imaging studies;
- Document operational, governance, and technical considerations to inform future scalable implementations.

As a result, this project will demonstrate novel technical architecture capabilities that are aligned with the perspectives of agency stakeholders:

- ARPA-H (funding agency): Demonstrate a federated approach to conducting analyses between local imaging sites, central imaging data aggregation, and study/cohort-specific analyses
- MIDRC (subaward organization): Support pipelines to link between public and private datasets via edge computing and preestablished connections at the site level through organizational partners such as ACR. MIDRC is a national initiative supporting the central collection of medical imaging data for research.

Collaborators/Roles

ACR – Grantee (ARPA-H subcontract issued through MIDRC). ACR is a mission-driven organization connecting members, academic partners, and industry leaders to accelerate scientific progress and transform patient care through the Center for Research and Innovation (CRI) and Data Science Institute (DSI). ACR has been issued a subaward to perform this work in accordance with an ARPA-H contract issued to MIDRC under the

Biomedical Data Fabric program. ACR is responsible for the overall conduct of this project and achievement of the defined aims during the course of the project period.

ACR is responsible for organizing a scientific research team that will conduct the project. This team will include investigators representing each participating site and ACR. ACR will engage with Sites to facilitate access to a breast cancer-related AI algorithm and transfer the aggregated results to ACR.

ACR Connect will be further developed to support medical image indexing and transfer of data generated from the Research project. ACR Connect will be made available to sites for installation within the institution firewall. AI algorithm results will be transferred to ACR via ACR Connect, along with the deidentified Medical Image Index (Appendix B) for each study.

ACR will interoperate with Datavant and Sites to facilitate the crosswalk of data between Veritas and Sites on patients who have completed breast cancer screening and have been lost to clinical followup. This will enable crosswalk of imaging metadata derived from the sites to be indexed centrally using ACR pseudo-IDs (Appendix A).

Veritas Mortality Data Linkage – Veritas is a provider of comprehensive U.S. mortality data, offering access to the “Fact of Death Mortality Index,” a comprehensive index of US mortality data from 1935 to present. The Index captures over 90% of deaths in the United States within two weeks of an individual’s death. It is the most complete, timely, and transparent fact of death solution on the market. Their index is designed to support timely and accurate identification of mortality outcomes for research and analytic purposes. In this project, Veritas serves as the external repository to evaluate the feasibility of privacy-preserving record linkage to connect to imaging datasets.

Sites – Participating Site collaborators will serve as data contributors and operational partners for the project. Site collaborators are responsible for identifying eligible cohorts and corresponding imaging datasets for indexing, running the AI algorithm on the cohort, and providing the results to ACR via ACR Connect for ACR’s aggregation and review.

Sites are responsible for ensuring appropriate local governance and compliance as well as alignment with the project plan.

Sites may also contribute to operational feedback regarding workflow implementation, resource considerations, and any regulatory, administrative, or technical constraints to inform future scalability. Specific research questions may be posed by the site’s research team based on the resulting data.

Datavant – Datavant is a leader in bringing data logistics to healthcare so organizations can securely and compliantly move health data from where it sits to where it needs to be.

Datavant will leverage the knowledge and presence it has gained through an ecosystem of deidentified data, footprint of identified data exchange, and expertise in patient privacy. They will facilitate data tokenization processes to support creating a secure linkage between imaging datasets at the site level and external data sources without the exchange of directly identifiable information.

MIDRC – Maintains centralized Imaging Index Library that serves as aggregation point for relevant study data/metadata on as needed basis. ACR will populate MIDRC/Gen3 with results of medical image indexing process for the cases included at the sites.

Approach

The approach for RADICAL is structured around two parallel and complementary workstreams: (1) demonstration of privacy-preserving record linkage between imaging data and external data sources, and (2) demonstration of the application of AI algorithms to imaging data to generate new analytic value in a federated environment. Each workstream can be executed independently while sharing common governance, documentation, and evaluation principles. For a detailed visual overview of these workstreams, see Appendix A.

Workstream 1 – PPRL: To demonstrate the feasibility of privacy-preserving record linkage between participating site imaging datasets and an external dataset, in this case mortality dataset, ACR proposes to collaborate with researchers at the participating sites who will be asked to create a patient cohort from breast cancer screening patients in accordance with project criteria as outlined below. Veritas will serve as the external data source and provide mortality data for linkage evaluation. Data preparation, eligibility determination, and local approvals remain under the control of each participating site. ACR does not directly receive, host, or process identifiable data as part of this workstream.

Two sites are being considered for project participation, University of North Carolina at Chapel Hill and University of Washington. Cohort creation will be initiated at the site level based on the specified date range of breast cancer screening study from July 2017 through December 2025. The cohort should contain a mix of cancer-positive and cancer negative patients across BIRADS categories, as well as at least 5 patients per site who were confirmed to be deceased whether from breast cancer or another mortality cause. Site-specific cohort requirements are as follows:

- At UNC, the additional requirement of including patients who were lost to follow up following participation in the TMIST study should be considered. The Tomosynthesis Mammographic Imaging Screening Trial (TMIST) is a large national study designed to evaluate breast cancer screening approaches and long-term outcomes. This trial is closed to accrual. As trial data remains protected, this proof of concept will

demonstrate the utility of linkage datasets on patients who have been lost to follow up following initial participation in the clinical trial. The definition of lost to follow up for cohort building purposes is as follows:

- For an annual participant in a screening round period, no information in over 1.5 years.
- For a biennial participant in a screening round period, no information in over 2.5 years.
- For a participant in long-term or cancer follow-up, no information in over 1.5 years.

Once the patient cohort has been defined, sites will identify associated breast imaging studies for further processing and analysis. ACR will provide sites with a medical imaging study indexing algorithm (see Appendix B) via ACR Connect, a HIPAA compliant platform utilized by ACR to facilitate data and image transfer. Sites will permit the algorithm to generate an index for each imaging study, and the resulting indices will then be transferred back to the ACR. Fully deidentified versions of the indices for each imaging study will be shared with MIDRC for inclusion in the national public library of indexed medical imaging studies hosted at Gen3, an open-source data platform maintained by the University of Chicago Center for Translational Science for building data commons and data ecosystems to further highlight the benefit of linking privately held data with a central repository for discovery purposes.

Following cohort identification and indexing, ACR Connect in cooperation with Datavant will initiate the tokenization process that will serve to link with the mortality data available in Veritas. Datavant will support execution of its PPRL process using its established methodology and infrastructure. Matches between patients at the Participating Site and deceased patients in the Veritas Index will be determined, and mortality information will be provided back to the institution for inclusion in clinical documentation as appropriate.

In addition to patient outcomes data, final outputs of this workstream will include a summary of feasibility observations and limitations and identification of site governance considerations, as well as a synthesis of lessons learned that could impact future scalability.

Workstream 2 - Federated AI: To demonstrate the value of a distributed AI network, this workstream will apply a breast imaging AI model to medical imaging data on prem and without central storage of images via federated technology provided through ACR Connect. Algorithm results on the cohort will be transferred to ACR via ACR Connect. The objective of this workstream is to demonstrate feasibility of deploying AI models at the edge and

FINAL_ Update w/ Mortality Data

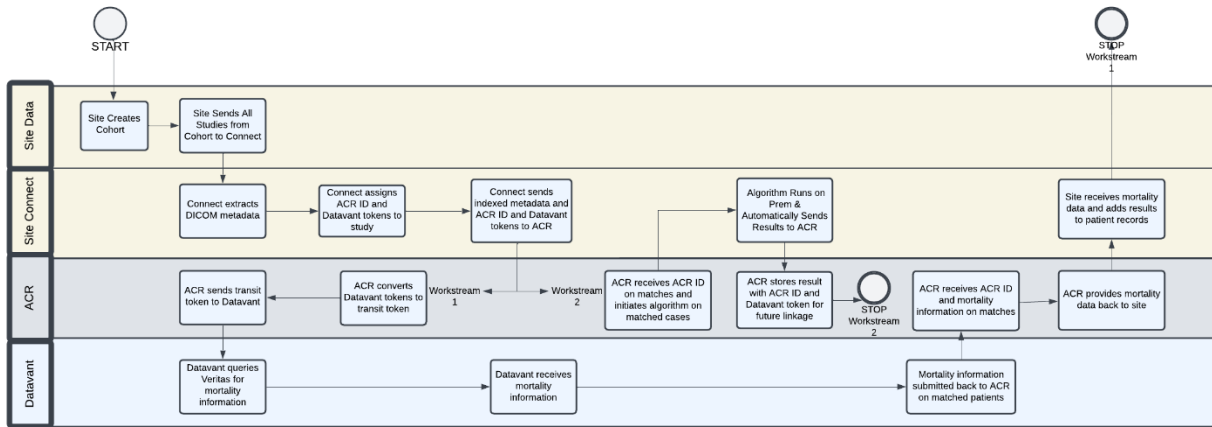
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aggregating model outputs without sharing patient identifiers, all within the broader context of enriching central datasets with AI results through cohort matching via PPRL.

Participating Sites will run the algorithm provided by ACR on the same cohort used in Workstream 1. Sites will apply the AI model using their local imaging data, and the algorithm output will be transferred to ACR via ACR Connect.

In the future, it is envisioned that as a result of this work, researchers will be able to assess model performance against subsequent images, outcomes data, and other genetic markers. For example, a researcher could investigate genomic profiles of patients rated to be at high risk of breast cancer and assess differences between patients that have been diagnosed with breast cancer within a defined timeframe and those that have not yet developed a breast cancer.

Appendix A – High Level Sample Workflow with Two Workstreams



Appendix B – ACR Medical Imaging Index schema

Level	Data Element
Program	Name
	Organ
Project	Focus Area
	Name
Facility	Name - Internal Attribute PHI
	ID
	Location - (State)
Patient	ACR ID
	Datavant ID
	Age
	Age Factor - (Years, Months or Days)
	Sex
	Race
	Ethnicity
	[Space within Schema for Future Use Case Tags]
Imaging Study	Study UID
	Accession Number - Internal Attribute PHI
	Year
	Patient Year of Birth
	Patient Sex
	Study Description
	[Space within Schema for Future Use Case Tags]
[Image series] – Common Elements – CT, MR, Nuclear	Manufacturer
	Manufacturer Model
	Software Version
	Modality
	[Sop Class]
	Body part Examined
	Contrast (y/n)
	Series UID
	Series Description
	[Space within Schema for Future Use Case Tags]
[MR Series] -	Magnetic Field Strength

FINAL_ Update w/ Mortality Data

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Unique Elements	Receive Coil Name
	Transmit Coil Name
	[Slice Thickness]
	[Scanning Sequence]
	[Sequence Variant]
	[Scan Options]
	[Acquisition Type]
	[Space within Schema for Future Use Case Tags]
Nuclear Series – Unique Elements	Radiopharmaceutical